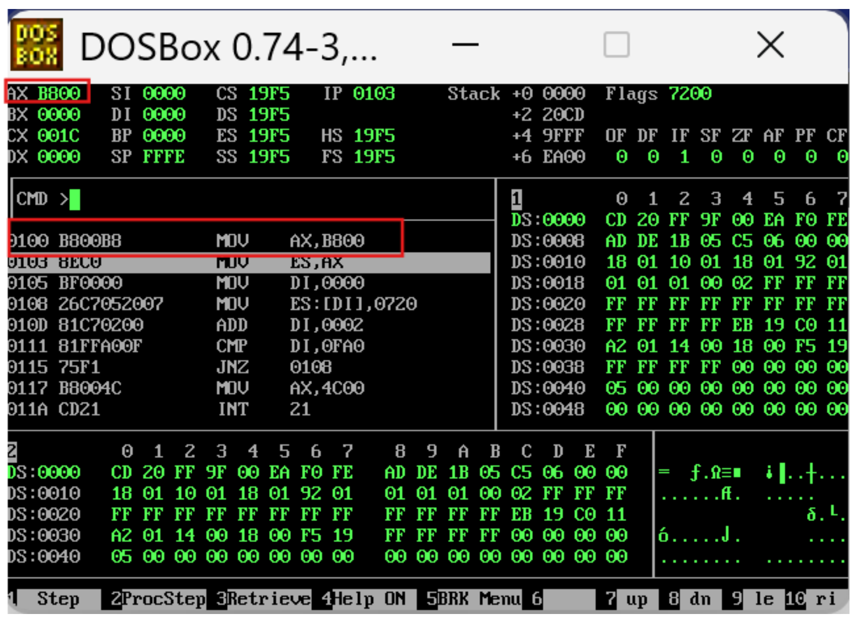
Name: Arsam Ali   Roll-No:23P-0038

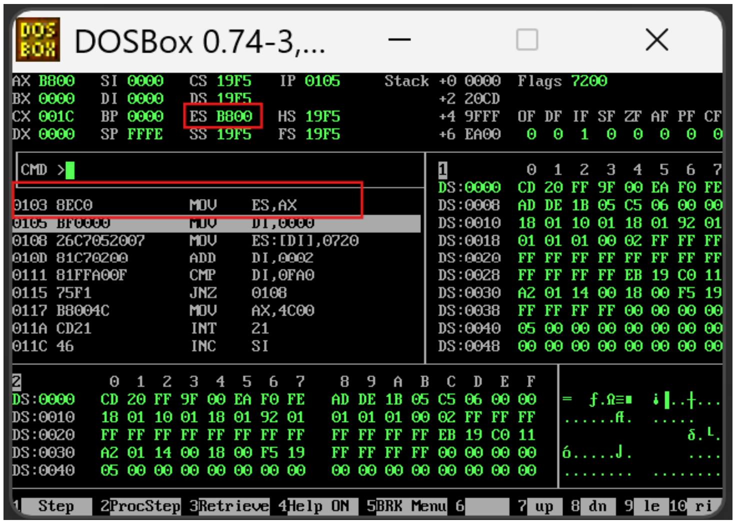
Code:

```
[org 0x0100]
            mov   ax, 0xb800
            mov   es, ax
            mov   di, 0
nextchar:   mov   word [es:di], 0x0720
            add   di, 2
            cmp   di, 4000
            jne   nextchar
            mov   ax, 0x4c00
            int   0x21
```

1) Address of memory address segment is loaded into the AX register

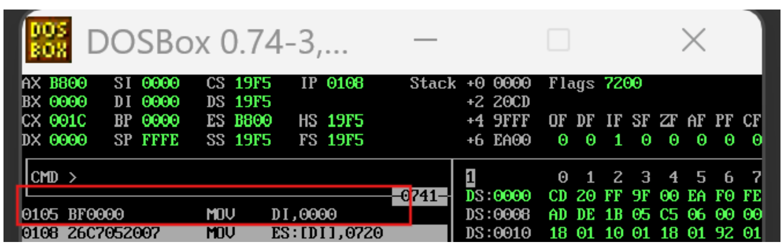


2) Value of AX (address of video address segment) is now stored in ES

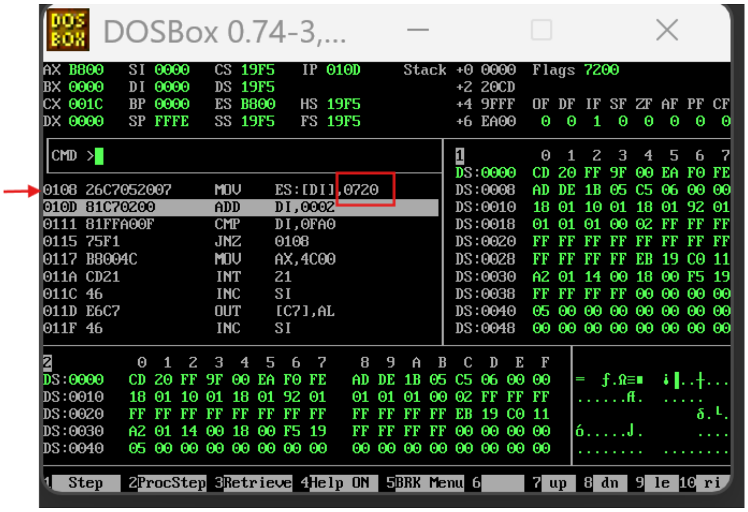


3) 0000 is moved in DI to make sure it stores no exitsting value and will be used as a offset

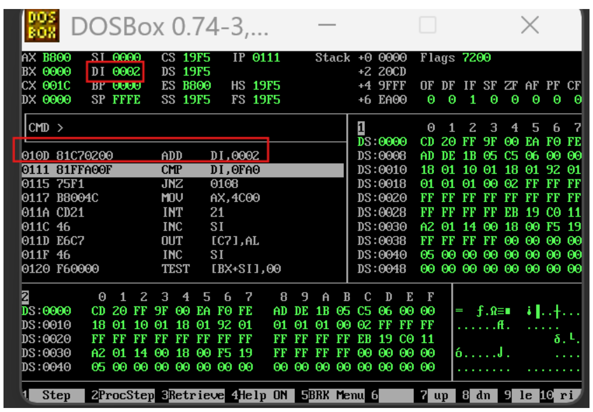
0000 now points to the first character on the screen



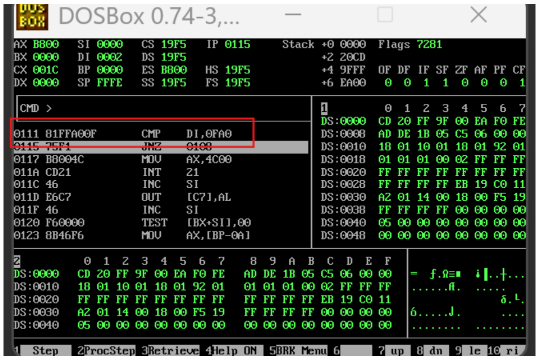
4) 0x720 represents attribute and character, where 0x07 represent attribute (black text on black background) and 0x20 represent character. So this writes a black space at the screen location.



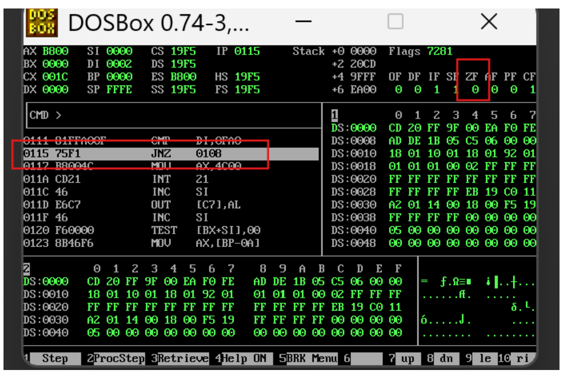
5) Adds 2 bytes in the offset as for each block (pixel) 1 for character and 1 for the attribute . so DI increments by 2 in each iteration



6) Screen has 80x25=2000 cells. Each cell = 2 bytes thus 2000 x 2 = 4000 bytes.
So when di is equal to 4000 (0FA0h), the entire screen is cleared.



7) This loop make sure the program completes all it iteration in this case 4000 if not it will repeat the process



--Program is terminated--